Preliminary.

Def. A Topological space X is called T_4 if:

For any disjoint closed sets A and B,
there exist disjoint open sets U and V such that $A \subset U$, $B \subset V$.

C-O Chain Lemma.

A space X is T_4 if and only if

For any closed C and open U with $C \subseteq U$,
there exists an open set O such that $C \subseteq O \subseteq \overline{O} \subseteq U$.

Proof. Suppose that X is T_4 space. Given closed C and open U,
$C \subseteq U$ implies $C \cap (X \setminus U) = \emptyset$. That is, C and $X \setminus U$ are disjoint closed set.
By T_4 condition, there exist disjoint open sets O, A such that

$$C \subseteq O, \quad X \setminus U \subseteq A, \quad O \cap A = \emptyset,$$

Now, $C \subseteq O \subseteq \overline{O}$, and $O \cap A = \emptyset \Leftrightarrow O \subseteq X \setminus A \Rightarrow \overline{O} \subseteq X \setminus A$
Since the closure is the smallest closed set containing them, so that
$X \setminus U \subseteq A \Leftrightarrow X \setminus A \subseteq U$ therefore

$$C \subseteq O \subseteq \overline{O} \subseteq U.$$

The converse can be shown for disjoint closed set A and B,
apply the chain for $A \subseteq X \setminus B$, similarly. ☐

Note: $A \subseteq B \Leftrightarrow B^c \subseteq A^c$, $A \cap B = \emptyset \Leftrightarrow A \subseteq B^c \Leftrightarrow B \subseteq A^c$.

[Def. Let X be a Topological space, and $A, B \subset X$ be disjoint closed sets
A real-valued continuous map $f: X \to [a,b]$ is called Urysohn function:

$$f|_A = a \quad \text{and} \quad f|_B = b$$

[Urysohn Lemma.
A T_4-space has an Urysohn function for any two disjoint closed sets.

Tietze Extension Theorem.
Let X be a T_4-space, and $A \subset X$ be a closed subset.
For any continuous map $f: A \to \mathbb{R}$, there exists a continuous map
$g: X \to \mathbb{R}$ such that $g|_A = f$.